Time Is the Baseline to Beat: Lead Time over a Timer as the Test for VLA Failure Monitors

Author Names Omitted for Anonymous Review. Paper-ID [add your ID here]

***Abstract*—On RoboCasa365, a failed vision-language-action (VLA) rollout is one that reaches its time limit. A runtime failure monitor therefore competes with a timer that alarms at a fixed fraction of that limit. Calibrated at a 5% false-positive rate (FPR), such a timer flags every one of 380 prospective failures, about 12 policy inferences before the timeout. Recall cannot separate a learned monitor from this timer; only lead time can. We test whether alarming when *either* a policy-internal score or the timer crosses its threshold warns earlier than time signals alone at a matched FPR, and calibrate this rule with a finite-sample FPR guarantee. On one VLA and 38 tasks, fusing the SAFE failure score with a single timer more than doubles the mean warning lead (0.090 to 0.207 of the time limit, exploratory), but most of that gain is task duration: a per-task timer, which alarms past the 95th percentile of each task's development success durations, nearly closes the gap without reading the policy. A preregistered test on a fresh cohort of 760 rollouts finds a small real gain over this per-task timer: 0.255 against 0.225 of the time limit (gain 0.030, 95% CI 0.007–0.093) at 0.061 FPR. Noise and within-task resampled scores fused the same way gain at most 0.009. The gain is above zero but below the margin of 0.05 we declared in advance, so the primary hypothesis is not supported. Stage 2 tests a second VLA, measures lead from the moment a failure becomes unrecoverable, and asks whether earlier alarms raise task success within a fixed time budget when they trigger a reset and retry.**

I. Introduction

Runtime monitors for VLA policies read the policy's hidden state and predict eventual failure [8], or score how unusual that state is relative to successful rollouts [27, 19]. They are compared with each other, but not with what the robot knows for free: how long the attempt has been running.

In RoboCasa365, as in most simulated manipulation benchmarks, an episode ends at first success or at a time limit, and in our test cohort every one of 380 failed rollouts ran to its limit. A timer at 90% of the limit therefore flags all of them at 4.7% FPR. Against it, a monitor can add only one thing: an earlier alarm at the same false-alarm rate, which leaves time to stop, reset or ask for help. We make this lead time the test, with three contributions.

1) **A testable hypothesis and endpoint.** We define the warning lead of an alarm as the fraction of the time limit left when it fires (zero for a missed failure) and state four hypotheses with decision rules fixed before the confirmatory data were analysed (Sec. IV).
2) **Calibrated OR-fusion with a guarantee.** We alarm when either a policy-internal score or the timer crosses its threshold, and calibrate both thresholds along a nested path with a split-conformal rule that bounds the FPR on a new successful rollout by α in finite samples (Proposition 1).
3) **Exploratory and confirmatory evidence.** SAFE fused with a single timer more than doubles its lead, but a per-task timer nearly matches it, and most failures are still recoverable at the limit (Sec. VI). On a fresh preregistered cohort the policy adds 0.030 of the limit over the per-task timer, below our 0.05 margin (Sec. VII).

The result is a small policy signal, not a useful one by our own standard: on timeout-defined benchmarks, policy internals add little to duration statistics, and monitors should be compared with a per-task timer and evaluated on failures that end early.

II. Related Work

Failure detection from policy internals. SAFE learns a failure score from a VLA's last-layer features and thresholds it with a functional conformal band [8]; we reuse its implementation. Follow-ups change the training data [18], labels [17, 14], features [28, 31] or policy [22]. None reports an elapsed-time detector. SAFE removes the duration shortcut by truncating rollouts to equal length and, on its real robot, by running every rollout to a fixed horizon [8]; a deployed monitor can do neither. Duration is a shortcut feature [7], but its causal form is legitimate online evidence.

Action-space and success-only detectors. STAC [1] and VLA-FAIL [20] compare overlapping action chunks, and ActProbe [9] feeds chunk consistency and magnitude to a task-conditioned LSTM, beating internal-feature baselines on RoboCasa. Our policy executes each chunk in full, so overlap signals are undefined. A task-conditioned recurrent detector can also learn how long each task takes, which our timers make explicit. FAIL-Detect [27] and FIPER [19] score inputs, outputs or embeddings against successful rollouts only. FIPER alarms when two conformal scores *both* exceed their thresholds, which suppresses false alarms; in its own pipeline an elapsed-time method matches its accuracy, but FIPER warns earlier at a matched false-alarm rate (Sec. VI). We alarm when *either* crosses, because the policy score and the timer catch failures at different times, and calibrate the pair jointly.

World-model verifiers and VLM monitors. Check-VLA [12] is the closest work on our benchmark. It verifies each action chunk with an action-conditioned world model on RoboCasa365, at a conformal 5% rate of unnecessary interventions, and repairs it with the same VLA. Foresight [24]

uses world-model latents. Neither compares against elapsed time; either could be fused with a timer. LLM and VLM monitors [1, 13, 21, 30, 6] are slower and complementary.

Time, calibration and intervention. VLA confidence drifts over a task [32]. Our time-risk baseline is a per-task survival-style estimate [10, 5], and lead time at a fixed false-alarm budget is the accuracy–earliness trade-off of early time-series classification [26]. Split-conformal thresholds give finite-sample false-positive control under exchangeability [3]; we extend the construction to a two-signal OR rule (Sec. V-C). RESCUE [2] finds that good detection scores do not guarantee lower intervention cost, and CheckVLA and FLARE [29] couple detection to repair; an earlier alarm matters only if acting on it raises success.

III. Problem Setting

Time. The simulator runs *control steps* at RoboCasa's default 20 Hz. The policy executes each 16-step action chunk in full, so *policy inferences* $t = 1, 2, \ldots$ (0.8 s each) are our clock, and a monitor decides only at an inference. Task k's limit of T_k control steps is $H_k = \lceil T_k/16 \rceil$ inferences (29 to 272, or 22.5 to 217.5 s). Every fraction of time (t/H_k, c, ℓ) is a fraction of H_k.

Monitoring. A monitor watches a rollout of a known task k and decides at each inference t whether to alarm, using only the policy features so far, t, the task and H_k. The first crossing is the alarm T (in inferences). FPR is the fraction of successful rollouts with any alarm. A rollout of length L inferences ends at success or at H_k; for a failure ($L = H_k$) the *warning lead* is $\ell = (L - T)/H_k$, or 0 if missed, and an alarm at the final inference counts with $\ell = 0$. The mean lead $\bar{\ell}$ combines recall and earliness. Detectors are compared at operating points fixed on a calibration cohort under FPR $\leq \alpha = 0.05$. Confidence intervals come from a paired bootstrap that resamples tasks, then rollouts within each task (2,000 replicates).

Scope. Whether elapsed time is a strong baseline depends on the termination rule. Where successes stop at completion (RoboCasa, FIPER's tasks, deployment), it is. Where every rollout runs to one fixed horizon, as in SAFE's real-robot set (in its release, 98% of failures and ≥ 30 successes per task stop at one step) or SimplerEnv, it is uninformative, and early-warning claims need failure-time annotations, which SAFE did not release.

IV. Hypotheses

We test four hypotheses. H1 and H2 were evaluated on a fresh confirmatory cohort with the decision rules below, registered as an amendment before any timer or lead quantity was computed on that cohort (Sec. VII); H3 and H4 are the Stage 2 extensions (Sec. VIII).

H1 (lead time, primary). At thresholds calibrated for FPR $\leq$ 0.05, OR-fusing a policy-internal score with the per-task timer (Sec. V), the strongest time-only monitor we know, increases the mean warning lead over that timer alone. *Supported* if the paired bootstrap 95% CI of $\Delta\bar{\ell}$ lies above zero, the point gain is at least 0.05 of the time limit, and the fused test FPR is at most 0.075 with a Wilson 95% upper bound at most 0.10. As a secondary check, the fused detector must also beat a timer whose fraction is tuned *on the test cohort* to the fused detector's test FPR, which favours the timer.

H2 (the gain needs policy evidence). Fusions that contain no policy-internal evidence (Gaussian noise, or SAFE scores resampled within task, each fused with the per-task timer) gain less than 0.05 of the time limit over the per-task timer.

H3 (generality). H1 holds for a second VLA with a different architecture.[TODO: Must change to match E2, which drops the 0.05 margin; author's call. Proposal: on RLDX-1's 20 tasks, (a) every failure runs to the time limit, (b) P leads the global timer, and (c) SAFE + P gains lead over P with a within-task 95% CI above zero, under the FPR rule of Sec. VIII.]

H4 (utility). When an alarm triggers a reset and retry, the OR-fused monitor yields a higher task success rate than the timer within the same per-task time budget.[TODO: Must change to match E3; author's call. Proposal: within a budget of 1.25 H_k, reset and retry triggered by P yields a higher success rate than no monitor; SAFE + P against P is descriptive.]

Only H1 and H2 are confirmatory; every other interval in this paper is descriptive. We expected H1 to fail (opened-data gain 0.007); it did, with a gain of 0.030 whose CI excludes zero (Sec. VII).

V. Method

A. Signals

Policy score. At each inference we take the final-layer action-token hidden state of the policy's action head (1,024 dimensions). The SAFE MLP [8] maps it to a failure score $u_t^{(j)}$ for each of three training seeds j. With per-task location $m_k^{(j)}$ and scale $\sigma_k^{(j)}$ estimated on development rollouts, the policy score is

$$s_t = \frac{1}{3}\sum_{j=1}^{3} \frac{\max_{t' \leq t} u_{t'}^{(j)} - m_k^{(j)}}{\sigma_k^{(j)}}. \tag{1}$$

The running maximum makes s_t monotone and does not change when a threshold is first crossed. Success-only alternatives (Mahalanobis [11], kNN [23], PCA-KMeans and RND [4]) are fitted on successful development rollouts and aggregated the same way.

Timers. The global timer alarms once $t \geq c\,H_k$ for a fraction $c \in (0, 1]$ shared by all tasks; it uses only the task's time limit. The per-task timer uses $c_k = \min(1, \rho_k + \delta)$, where ρ_k is the 95th percentile of task k's development success durations as fractions of H_k and the shared margin δ is set by Proposition 1 on calibration successes. It needs successful rollouts of each task, but no failures and no policy features. We fixed the 95th percentile after auditing the opened cohort, so we report 0.75, 0.90 and 1.0 (the longest success) as sensitivity.

Time risk (baseline). The survival-style alternative estimates $r_t = \Pr(\text{failure} \mid k, \text{still running at } t)$ from development rollouts: with n_t of them still running at t and f_t of those

failing, it fits an increasing isotonic regression in t/H_k to $(f_t + 0.5)/(n_t + 1)$, weighted by n_t. Because every failure runs to H_k, r_t is a task-specific timer; its smoothing keeps it below a high threshold in tasks with few development failures, which is why it misses failures the plain timer catches.

B. OR-fusion

The fused monitor alarms at

$$T = \min\{\, t : s_t > \tau_s \text{ or } t \geq c\, H_k \,\}. \tag{2}$$

Neither signal is restricted in time: the policy score may alarm early, and the timer guarantees an alarm on every failure, at $c\, H_k$ at the latest. The lead gain over the timer comes only from failures on which the policy score crosses first.

C. Calibration with a finite-sample FPR guarantee

A successful rollout i of length L_i (in inferences) raises a false alarm under (τ_s, c) if and only if $M_i = \max_{t \leq L_i} s_t > \tau_s$ or $\rho_i = L_i/H_{k_i} \geq c$, so its false alarm depends only on the pair (M_i, ρ_i). Fix in advance a path $\gamma(\lambda) = (\tau_s(\lambda), c(\lambda))$, $\lambda \in [0, 1]$, with both coordinates continuous and nondecreasing, so that a larger λ is more conservative and $\gamma(1)$ never alarms on a success. For each calibration success define

$$V_i = \min\{\lambda : M_i \leq \tau_s(\lambda) \text{ and } \rho_i < c(\lambda)\}, \tag{3}$$

the least conservative point on the path at which rollout i stays silent. Because the path is monotone, rollout i is silent at λ exactly when $\lambda \geq V_i$.[1]

Proposition 1: Let n calibration successes and a new successful rollout be exchangeable, and let the path γ not depend on the calibration successes. Set $\hat{\lambda} = V_{(\lceil (n+1)(1-\alpha) \rceil)}$, the $\lceil (n+1)(1-\alpha) \rceil$-th smallest V_i (or $\hat{\lambda} = 1$ if that rank exceeds n). Then the OR-fused monitor at $\gamma(\hat{\lambda})$ raises a false alarm on the new success with probability at most α.

Proof: The new rollout alarms at $\hat{\lambda}$ if and only if $V_{n+1} > \hat{\lambda} = V_{(r)}$ with $r = \lceil (n+1)(1-\alpha) \rceil$. By exchangeability, the rank of V_{n+1} among $V_1, \ldots, V_{n+1}$ is uniform on $\{1, \ldots, n+1\}$ (ties broken at random), so $\Pr(V_{n+1} > V_{(r)}) \leq 1 - r/(n+1) \leq \alpha$; this is the split-conformal quantile lemma [3] applied to the scalar V. ∎

The path may use any data except calibration successes, such as development rollouts or calibration *failures*. The guarantee is marginal over tasks; a per-task version needs 19 calibration successes per task.

The preliminary results use the original protocol's joint search: among all pairs of calibration maxima with calibration FPR $\leq \alpha$, keep the one with the largest mean lead (highest recall for time-risk fusions). This best-of-many selection on 114 successes has no guarantee and is optimistic (test FPR 0.058–0.068 against 0.044 on calibration). The per-task timer rows, and the confirmatory test, use Proposition 1 with the path $\delta = \lambda$, $\tau_s = u + s\lambda$ and (u, s) chosen on calibration failures.

[1]The strict $\rho_i < c(\lambda)$ makes the set right-open; taking c on a finite grid, as the timer's resolution of one inference does anyway, makes the minimum attained.

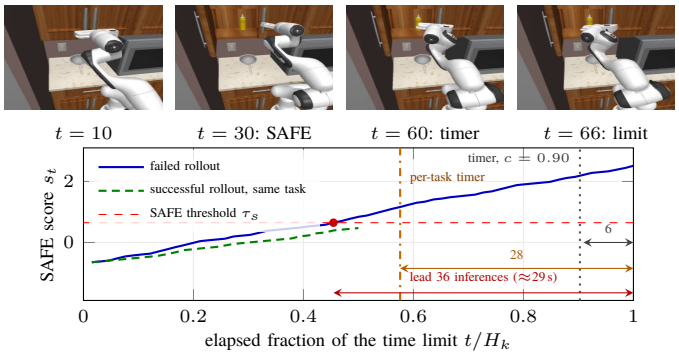


Fig. 1. One failed *OpenCabinet* rollout from the prospective test (chosen to illustrate; the median SAFE-first lead is smaller). The SAFE score crosses its calibrated threshold at $t = 30$ of $H_k = 66$ inferences, 36 inferences before the time limit; the timer alarms at $t = 60$, and the per-task timer, which never reads the policy, at $t = 38$. Dashed: the longest of the 9 (out of 10) successful test rollouts of this task that stay below the threshold; it ends at $t = 33$.

TABLE I

PROSPECTIVE TEST (380 SUCCESSES, 380 FAILURES), THRESHOLDS FROM CALIBRATION AT FPR ≤ 0.05. $\bar{\ell}$: MEAN LEAD (MISS = 0); MED: MEDIAN LEAD (INFERENCES) OF DETECTED FAILURES; @.75: FAILURES ALARMED BY $0.75\, H_k$. *MEAN OF 20 DRAWS ($\bar{\ell}$ 0.088–0.134). †MARGIN BY PROPOSITION 1.

Detector	Recall	FPR	$\bar{\ell}$	Med	@.75
Timer ($c = 0.903$)	1.000	0.047	0.090	11.5	0.000
Time risk	0.632	0.047	0.120	16.5	0.211
Time risk + timer	1.000	0.050	0.125	11.0	0.105
Noise + timer*	1.000	0.066	0.101	–	–
SAFE only	0.566	0.039	0.190	23.0	0.421
SAFE + time risk	0.955	0.058	0.184	12.0	0.311
Per-task timer†	1.000	0.045	0.195	13.5	0.342
SAFE + per-task timer†	1.000	0.042	0.202	14.0	0.389
SAFE + timer	1.000	0.058	0.207	10.0	0.421

VI. PRELIMINARY EVIDENCE

Setup. We use the released Xiaomi-Robotics-1 RoboCasa365 checkpoint [25], which executes each 16-step action chunk in full, on the 38 official RoboCasa365 tasks [15, 16] with published success between 5 and 45 of 50 and enough failures (10 atomic, 28 composite; failures last 29–272 inferences). Disjoint cohorts serve one purpose each: development (1,824 rollouts) fits SAFE and the time risk; calibration (3 successes and 3 failures per task; 454 rollouts including those past the quota) sets thresholds; the prospective test (10 per class per task) is evaluated once; a confirmatory cohort of the same size is used only in Sec. VII. SAFE, its normalization and the time risk were frozen before the test was collected; the timers and fused detectors were added later, with thresholds from calibration only, so the results below are exploratory.

The timer is the baseline to beat. The timer ($c = 0.903$) alarms on all 380 test failures at 0.047 FPR with a fixed lead of 0.090 (Table I). The time-risk curve misses 140 failures, all in 14 tasks whose smoothed curve never reaches its threshold, so recall gains measured against it (to 0.90–0.96 for SAFE and four success-only scores) vanish here.

SAFE adds lead over a single timer. SAFE + timer raises the mean lead to 0.207 (paired gain 0.117, CI 0.060–0.178;

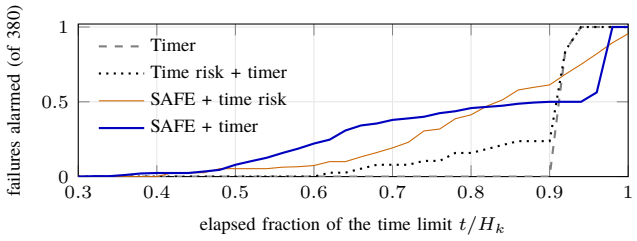


Fig. 2. Fraction of the 380 test failures alarmed by each elapsed fraction of the time limit. SAFE + timer catches half of the failures early and the rest at its timer fallback ($c = 0.96$).

TABLE II

CONFIRMATORY TEST (380 SUCCESSES, 380 FAILURES). P = PER-TASK TIMER ($q = 0.95$). H1 INTERVAL: REGISTERED DECISION BOOTSTRAP; OTHERS: TEST ONLY. *MEAN OF 20 DRAWS [MAXIMUM]. †FROZEN ORIGINAL THRESHOLDS.

Detector	Recall	FPR	$\bar{\ell}$	Gain vs P
Global timer	1.000	0.024	0.123	−0.102 (−0.147, −0.059)
SAFE + time risk†	0.955	0.039	0.190	−0.035 (−0.066, −0.007)
SAFE + global timer	1.000	0.047	0.215	−0.011 (−0.068, 0.045)
P	1.000	0.055	0.225	–
Noise + P*	–	–	–	0.002 [0.009]
Resampled SAFE + P*	–	–	–	0.002 [0.007]
SAFE + P (H1)	1.000	0.061	**0.255**	**0.030** (0.007, 0.093)

20.2 against 11.7 inferences) and flags 42% of failures by $0.75\,H_k$ (Figs. 1 and 2), at 0.058 test FPR. Noise fused with the timer reaches 0.101, and the time risk with the timer 0.125. Among rollouts still running at 10, 25, 50 and 75% of the limit, SAFE's within-task ROC-AUC is 0.55–0.60, against 0.41–0.51 for noise.

A per-task timer removes the gain (H1, exploratory). The per-task timer reaches $\bar{\ell} = 0.195$ at 0.045 FPR without reading the policy. SAFE + timer leads it by an insignificant 0.012 (CI -0.044 to 0.063). SAFE + per-task timer reaches 0.202, a gain of 0.007 (CI -0.003 to 0.019), far below the 0.05 margin of H1. The two effects cancel across tasks: SAFE + timer leads where the time risk never fires (0.405 against 0.221), and trails elsewhere.

No early warning, and mostly no early failure (exploratory). No detector flags over 8% of failures by half the limit. Is timeout-defined failure then trivial by construction? One labeler, blind to alarms, marked the point of no return (after which the policy could not finish in time without help) on 48 confirmatory-cohort failures (2 per atomic, 1 per composite task): 38 (79%) were still recoverable at the limit (atomic 17/20, composite 21/28), most often slow or nearly successful. A second labeler agreed on 11 of 12 (Cohen's $\kappa = 0.80$). Failure here mostly means not finishing in time, which is why elapsed time is so informative; an early "failure" alarm mostly predicts slowness. On the 10 decided failures, P alarmed after the point of no return on all, SAFE + P at or before it on 2 of 3 atomic and 0 of 7 composite ones (an illustration).

A second policy, offline (RLDX-1, exploratory). On existing RLDX-1 data (20 tasks; 740 development rollouts, 60 held-out calibration successes, 100/100 test rollouts), all 438 failures ran to the time limit. P reaches $\bar{\ell} = 0.271$ at 0.03 test FPR, the global timer 0.032. SAFE (LSTM head) + P gains 0.034 over P (CI 0.008–0.067) at the *same* FPR, so unlike on XR-1 none of it is extra FPR budget; the MLP head adds nothing. The gain is atomic (0.049, CI 0.013–0.097); on composite tasks it is 0, as SAFE never alarms before P. Caveats: the fused path was chosen on in-sample development failures, the FPR margin rests on 60 successes, and noise + P matched the gain in 2 of 20 draws, both at 0.08 FPR.

The test on a published benchmark (exploratory). In FIPER's released rollouts [19] (five tasks, two on a real robot), length alone separates failures with ROC-AUC 0.92–1.00 in four tasks (0.62 in the fifth). We re-ran FIPER's own code, which reproduces its published results within 0.05, adding one method whose score is elapsed time. Under FIPER's thresholds and selection rule, the timer has the best balanced accuracy (0.81 against 0.78) and a test FPR of 0.09, against 0.35 for FIPER. At FIPER's FPR, however, FIPER warns 0.146 of the episode earlier than the timer (CI 0.104–0.188; post hoc), on four of five tasks. The test thus credits FIPER, but finds SAFE's gain on XR-1 small (Sec. VII): it discriminates both ways.

VII. CONFIRMATORY TEST OF H1 AND H2

Protocol. A fresh test cohort (new seeds, 10 per class per task) was collected with SAFE frozen and preregistered for recall against the time risk. Before any timer or lead was computed on it, but with its recall and FPR known, an amendment made the per-task timer P ($q = 0.95$) the comparator and $\Delta\bar{\ell}$ the endpoint, with the rules of Sec. IV. A new calibration cohort of the same design (380 successes) sets the thresholds of P and SAFE + P by Proposition 1. The decision interval resamples both cohorts and refits thresholds in each of 2,000 replicates. Every test failure again ran to H_k.

H1 is not supported, but the gain is not zero. SAFE + P leads P by 0.030 of the time limit (Table II). Its decision CI lies above zero and its test FPR (0.061, Wilson 95% upper bound 0.089) meets the FPR rules, but the point gain is below the 0.05 margin, so H1 fails. The CI (upper end 0.093) does not show the gain is negligible, and the opened-cohort estimate (0.007) understated it.

H2 is supported. Noise and within-task resampled SAFE scores, fused with P and calibrated the same way, gain at most 0.009 over P in 20 draws each. What SAFE adds is policy evidence, not an artifact of fusing any score with a timer.

Task duration carries most of the lead. P alone leads the global timer by 0.102, about what SAFE + global timer reaches, and the original SAFE + time-risk design, which beat the time risk on recall here, gives 0.035 *less* lead than P. Part of the H1 gain is FPR budget: a per-task timer tuned on the test cohort to SAFE + P's FPR leaves a gain of only 0.015 (CI -0.007 to 0.041), and a global timer tuned the same way 0.024 (CI -0.022 to 0.074), so the secondary check of H1 also

fails. The gain is 0.026–0.030 for $q = 0.75$–0.95 and 0.074 at $q = 1$, where P itself is weaker (0.164).

Atomic tasks carry the gain (descriptive). On the 10 atomic tasks SAFE + P gains 0.081 (0.372 against 0.291); on the 28 composite tasks, 0.012. Exploratory, on the same cohorts: the four frozen success-only scores fused with P gain 0.020–0.034 (SAFE 0.030), so on lead, as on recall, the choice of score matters little; and all five gain more on atomic (0.045–0.081) than on composite tasks (0.008–0.027). The split is a property of atomic tasks under the per-task timer, not of SAFE, and RLDX-1 shows it too. On composite tasks P likely lacks progress: on five, an oracle reading subtask predicates gained 0.071 over P (task-level CI -0.021 to 0.129). FPR is not controlled per stratum.

VIII. Stage 2 Plan

E2: Second policy (H3). RLDX-1 on its 20 tasks (Sec. VI), fresh calibration and test cohorts of 20 successes and 20 failures per task (about 2,500 rollouts), with held-out calibration failures. Primary: SAFE + P against P, superiority (decision 95% CI above zero), conditional on these 20 tasks (rollouts resampled within task), under the FPR rule below. Offline sizing gives power 0.92 at the offline gain of 0.034 (the CI excludes zero in every simulated study, so the FPR rule decides); a 0.05 margin would have power ≤ 0.09, so we drop it. Secondary, no power claim: the CI with tasks resampled (power about 0.54). Descriptive stratum: the 14 atomic tasks. It replaces an atomic-only XR-1 test (power ≤ 0.55 at the 0.05 margin up to 60 per class per task; its 0.089 estimate is optimistic, about 0.06 with all retained successes). With 5 held-out failures per task, all sizing figures are optimistic.

FPR rule.[TODO: Proposed rule; author to confirm.] Calibration targets $\alpha = 0.05$. H1's fixed limit (FPR ≤ 0.075, Wilson bound ≤ 0.10) rejects good detectors in small cohorts; instead, false alarms must not reject "true FPR ≤ 0.075" in a one-sided exact binomial test at level 0.05: at most the 95th percentile of Binomial(n_s, 0.075) for n_s test successes (39 of 400 in E2). This is looser, so E2's power stays ≥ 0.92.

E3: Reset and retry (H4). On XR-1's 38 tasks[PENDING: E3 policy not chosen; XR-1 assumed (RLDX-1: about 480 per arm)], an alarm ends the attempt and resets the scene while a budget of $1.25\,H_k$ remains. Registered: P against no monitor, paired by seed (about 650 episodes per arm). SAFE + P is descriptive: separating it from P needs thousands to over a million episodes per arm (about 1,450 on XR-1 atomic tasks at $1.0\,H_k$, an optimistic bound). As most failures are still recoverable at the limit, a "more time, no reset" arm (to $1.25\,H_k$) is the natural comparator.[PENDING: sample size of the no-reset arm]

E4: Checks. Action-only monitors (chunk magnitude, second-sample disagreement); richer time baselines (end-effector stall, discrete-time hazard); OR vs AND on full curves; SAFE faithfulness; precision at deployment prevalence; unseen tasks. **E5: Failures decided early.** E5 targets the minority of failures with an early point of no return, with several labelers, a written protocol and adjudication (our two labelers' timings differed by 0.24–0.34 H_k), and reports lead from it.

IX. Limitations

All results are simulated shadow evaluations (monitors never intervened), from one policy prospectively and a second (RLDX-1) only offline, on existing data with a small test set. Timers and fused detectors were defined after the exploratory cohort was opened, and the confirmatory endpoint after that cohort's recall and FPR were known. Timers and normalization assume a known task and limit; the policy score needs white-box access. Test cohorts are balanced by quota, so recall and FPR are per class, not at deployment prevalence. Point-of-no-return labels are one labeler's video judgments, checked on 12 videos; only 10 failures were decided. The FIPER lead comparison is post hoc, with 10 + 10 test rollouts per real-robot task (PushChair: a lower bound).

X. Conclusion

When failure means timeout, and mostly slowness, a learned monitor must warn earlier than a matched-FPR timer, as FIPER does on its benchmark. A per-task timer nearly matched a hidden-state score fused with a single timer; on a fresh preregistered cohort the hidden state added 0.030 of the time limit, below our declared margin. Stage 2 tests a second policy and closed-loop retry.

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